When Does Non-Myopic Bayesian Experimental Design Work with Language Models?

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Abstract

Sequential Bayesian experimental design (BED) offers a principled way for language-model agents to ask questions, update beliefs, and plan future information gathering. Yet deeper planning can optimize errors in an LLM-derived probabilistic model rather than useful information. We audit this failure across an exact constrained-location test, semantic 20 Questions, and two external interactive benchmarks: Paprika troubleshooting and MediQ clinical reasoning. Depth-two planning succeeds in the exact test, while one-step EIG substantially beats prompting in 20 Questions. We then test StrategyEIG: an LLM proposes short contingent policies and an exact Bayesian verifier selects their root. Across five preregistered Rock Diagnosis maps (3–15 rocks), StrategyEIG improves entropy AUC over the same LLM roots scored myopically by 0.475–0.941, exhaustive one-step width by 0.452–0.939, and matched random strategies by 0.362–0.887; all fifteen paired 95% intervals exclude zero and truth-log-posterior AUC agrees. GPT-5.4 Mini replicates the 8-, 11-, and 15-rock gains. The selected policies move to enable accurate future checks while both myopic controls never move; exact and direct-vLLM gates at 26B and 12B preserve the gap at 32,768 latent states, while E4B fails. A successor-grounded 26B gate recovers exact d3 on 16/16 range-gated cells, but a tied-control identity audit blocks trajectories. Two data-derived semantic unlock tasks reproduce exact d2 gains; a UCI Thyroid task with native delayed-assay metadata further passes a 26B proposal gate. Names-only transfer fails, but calibrated utility cards recover 100% of exact d2 on UCI Mushroom with zero projection and 99.4% on Thyroid with bounded projection, beating d1 and random. Cleveland similarly recovers 100.9% with zero projection and positive entropy AUC, although independent truth-log corroboration against random is marginal. Negative location, Paprika, and MediQ audits identify the boundary: depth needs both a deployment-matched generative model and verified non-greedy structure. The result supports an LLM-Modulo architecture for sequential BED—semantic policy generation coupled to exact external verification—rather than unverified LLM simulation alone.

1 Introduction

Interactive agents must often acquire information before committing to an answer or action. A clinical agent asks about symptoms; a support agent diagnoses a fault; a guessing agent narrows a large semantic space. Direct prompting is an unreliable solution: MediQ reports that asking models to interact can reduce accuracy relative to answering from the initial context [Li et al., 2024]. Bayesian experimental design instead selects a query by its expected information gain (EIG) about a target variable. BED-LLM makes this practical in semantic spaces by using an LLM to propose hypotheses and provide response likelihoods [Choudhury et al., 2026].

The natural next step is non-myopic BED: score a query by both its immediate EIG and the information available to a future adaptive query. Recent methods simulate conversation trees for this purpose [Hu et al., 2024, Arnould et al., 2026]. In principle this can value a prerequisite question with little immediate gain. In practice every additional branch invokes the same uncertain model, changes the belief on which later calls condition, and maximizes over more noisy continuation scores. A larger tree can therefore make a confidently worse decision.

We investigate this distinction using a validation-first sequence of experiments. An exact constrained-location environment verifies the recursion. Rock Diagnosis then supplies the missing combination of an exact simulator, enabling movement, LLM-generated branch policies, and external exact verification. Semantic 20 Questions verifies one-step transfer but does not establish an incremental horizon benefit. Natural location experiments show non-monotonic depth under LLM-generated strategies. Finally, Paprika and MediQ expose different failures on external benchmarks: task utility and endpoint attribution in Paprika, and latent-state sufficiency plus record sparsity in MediQ.

Our contributions are:

1. StrategyEIG, an LLM-Modulo sequential BED architecture in which an LLM proposes explicit contingent policies and an exact Bayesian component verifies their total information gain [Kambhampati et al., 2024];
2. preregistered paired evidence that both the LLM proposal prior and non-myopic verification are load-bearing in a depth-favorable Rock geometry;
3. a cross-environment audit that separates correctness of the planning recursion from validity of the probabilistic interaction model;
4. two external-benchmark autopsies with invalid outcomes quarantined rather than repurposed as method evidence; and
5. a concrete validation contract for future non-myopic LLM BED: target and latent sufficiency, calibrated likelihoods, branch/deployment equivalence, a verified greedy gap, and ranking fidelity before policy evaluation.

2 The Probabilistic Contract

Let $h_t = ((x_1, y_1), \ldots, (x_t, y_t))$ be the interaction history and θ the finite target decoded at the endpoint. A generative state z may contain additional information needed to predict responses, with $\theta = g(z)$. The relevant model is

$$p_t(\theta, z)\, p(y \mid z, x, h_t), \qquad p(y \mid \theta, x, h_t) = \sum_z p(y \mid z, x, h_t) p_t(z \mid \theta).$$

When θ itself is sufficient, as for a hidden animal answering a property question, z can be omitted. When θ is merely a decision label, such as "treat hypoperfusion first," it does not determine a patient's glucose or history and cannot serve as the response-generating state.

For a coherent model, one-step EIG is

$$I_t(x) = \sum_{\theta,y} p_t(\theta)p(y \mid \theta, x, h_t) \log \frac{p(y \mid \theta, x, h_t)}{\sum_{\theta'} p_t(\theta')p(y \mid \theta', x, h_t)}.$$

Our exact depth-two score is

$$Q_2(x) = I_t(x) + \mathbb{E}_{y|x,h_t} \left[\max_{x' \in \mathcal{X}(h_t,x,y)} I_{t+1}(x') \right].$$

It branches over every positive-mass response, performs a Bayesian update on the fixed target support, generates branch-specific follow-ups, and adds the best second-step EIG. This is mathematically appropriate for the model supplied to it. It cannot determine whether that model describes deployment.

Endpoint target θ → sufficient state z → calibrated $p(y \mid z, x, h)$ → matched simulated/deployed update → verified greedy gap → rankable depth increment → endpoint comparison

Figure 1: The validation chain for non-myopic LLM BED. Failure of any upstream link invalidates evidence from downstream policy performance.

Two requirements are therefore distinct. *Model validity* asks whether the target, latent state, likelihood, update, and simulated branches form one process that matches deployment. *Planning value* asks whether the task contains complementarity or gating such that a lower-immediate-EIG query enables a more informative continuation. If information has adaptive diminishing returns, greedy policies can already be competitive [Golovin and Krause, 2011]. Depth needs a demonstrated gap, not merely a multi-turn interface.

3 Audit Design and Evidence Rules

We aggregate only preregistered or reconstructable paired evidence. Exact location tests use the known source prior and signal likelihood. The animals audit reconstructs every 20-round binary trial and reports deterministic bootstrap intervals. Paprika and MediQ use hash-pinned official releases and strict transcript audits. Failed parsing, unsupported simulator facts, and endpoint contradictions fail closed.

Most importantly, we distinguish *policy evidence* from *diagnostic evidence*. A Paprika run whose customer reports the private correct remedy as failing cannot compare policies, even if aggregate resolution numbers exist. A MediQ run with an invalid likelihood cannot establish EIG value, even if its queries are fluent. Such runs may localize a broken link in Figure 1; they do not support an accuracy claim.

Setting	Generative model	Greedy gap	Key result	Evidence status
Exact constrained location	Exact	Explicit delayed trap	RMSE: d2 .153, greedy .561	Positive control
Rock Diagnosis	Exact finite state	Enabling movement	Strategy d2 beats d1, width, random	Paired positive
Range-gated Rock	Exact finite state	Two-step travel	26B h3 proposals recover d3 on 16/16	Proposal positive; no trajectory
Mushroom / Cleveland / Thyroid	Exact empirical	Zero-EIG acquisition/workup	Grounded GPT recovers 100/.994/1.009 of d2	Positive / partial
Animals 20Q	Concrete target, binary replies	Unverified	d1 AUC .743, think-naive .465	One-step positive
Natural location	Exact signal, LLM strategies	Weak/variable	d1/d3/d5 non-monotonic	Horizon negative
Paprika	Insufficient state/task utility	Assumed, not enforced	Endpoint contradiction	Diagnostic only
MediQ iMEDQA	A–D often not generative	Channel too sparse	8/30 replies available	Gate failure

Table 1: The same recursion behaves differently as the probabilistic and task contracts weaken. "d1" and "d2" denote scoring horizon.

4 Results Across Environments

Exact lookahead can work.

We first constructed a branch-decoy source prior with a local-bump observation model and movement-constrained queries. The first query can reveal which region to enter, while movement limits make that choice affect later measurements. Over 120 paired trials, exact depth two reduced final RMSE from 0.5606 for greedy EIG to 0.1529, a mean paired difference of -0.4077. This is a harness result, not an external-benchmark claim. It shows that the implementation and objective can exploit a genuine delayed-information structure when model error is absent.

One-step semantic BED transfers; depth does not yet.

On animals 20 Questions, depth-one EIG achieved accuracy AUC 0.743 versus 0.465 for a thinking naive policy and 0.458 for non-thinking naive over 40 paired trials. The AUC gain over thinking naive was 0.278 with 95% bootstrap interval $[0.191, 0.365]$. This is the stable positive BED result.

The apparent aggregate depth-two AUC was 0.770, but the full depth-one and depth-two streams were not paired. On the only exact ten-trial seed block, depth two minus depth one was -0.040 with interval $[-0.105, 0.025]$ and 3/1/6 wins/ties/losses.

Depth three was worse than depth two across 40 paired trials: -0.126 $[-0.224, -0.035]$. Thus these data support semantic EIG, not non-myopic improvement.

Natural location experiments reinforce this distinction. In a 30-trial movement-constrained study using exact physical likelihoods but LLM-generated strategies, final RMSE was 0.412 for greedy EIG and 0.929, 1.127, and 0.542 for strategy horizons 1, 3, and 5. Without the movement constraint all horizons were effectively flat. More planning did not yield monotonic behavior, and the naive policy remained competitive.

LLM proposals plus exact lookahead work in enabling geometry.

The negative natural-location sweep did not test the architecture where depth itself was useful: a later exact width gate showed that its smooth sensing geometry favored one-step action coverage. We therefore moved to finite Rock Diagnosis. Movement has exactly zero immediate EIG, but changes distance-dependent sensor accuracy. The LLM returned six explicit two-action branch policies per decision: three distinct movement roots followed by checks, and three direct-check roots. A finite exact posterior and exhaustive observation branching scored every policy; the LLM was never called inside rollout scoring.

We preregistered 30 paired eight-round trajectories on each of two paper maps, then scale extensions on the canonical eight-rock RockSample[7,8] geometry from Smith and Simmons [2004], the standard eleven-rock RockSample[11,11] SARSOP instance, and a frozen fifteen-rock RockSample[15,15] POBAX instance. This is a diagnosis-only adaptation: it retains the latent rock vector, grid movement, and distance-dependent sensor, while excluding sampling, reward, and exit actions. Controls remove one component at a time: shared d1 scores the identical LLM roots myopically; width uses exhaustive d1 over all legal actions with matched exact-scorer units and one LLM ordering call; random strategy uses the same branch grammar and exact d2 verifier with random legal policies. Table 2 reports paired gains in mean posterior-entropy AUC; positive is better for StrategyEIG.

Map	Control	Entropy-AUC gain	95% CI	W/T/L
3–6	Shared-roots d1	.5205	[.4857, .5531]	30/0/0
3–6	Exhaustive d1 width	.5205	[.4855, .5537]	30/0/0
3–6	Random strategies	.3748	[.3229, .4267]	30/0/0
5–7	Shared-roots d1	.4751	[.3929, .5544]	30/0/0
5–7	Exhaustive d1 width	.4517	[.3677, .5310]	29/0/1
5–7	Random strategies	.3622	[.2714, .4482]	29/0/1
7–8	Shared-roots d1	.7502	[.7050, .7938]	30/0/0
7–8	Exhaustive d1 width	.7297	[.6874, .7724]	30/0/0
7–8	Random strategies	.6735	[.6251, .7212]	30/0/0
11–11	Shared-roots d1	.9412	[.8968, .9807]	30/0/0
11–11	Exhaustive d1 width	.9387	[.8952, .9773]	30/0/0
11–11	Random strategies	.8867	[.8305, .9385]	30/0/0
15–15	Shared-roots d1	.6723	[.6328, .7124]	30/0/0
15–15	Exhaustive d1 width	.6700	[.6301, .7111]	30/0/0
15–15	Random strategies	.6345	[.5722, .6877]	30/0/0

Table 2: Preregistered StrategyEIG confirmation. All fifteen primary paired bootstrap intervals exclude zero. Truth-log-posterior AUC also favors StrategyEIG in all fifteen comparisons, with all fifteen intervals excluding zero.

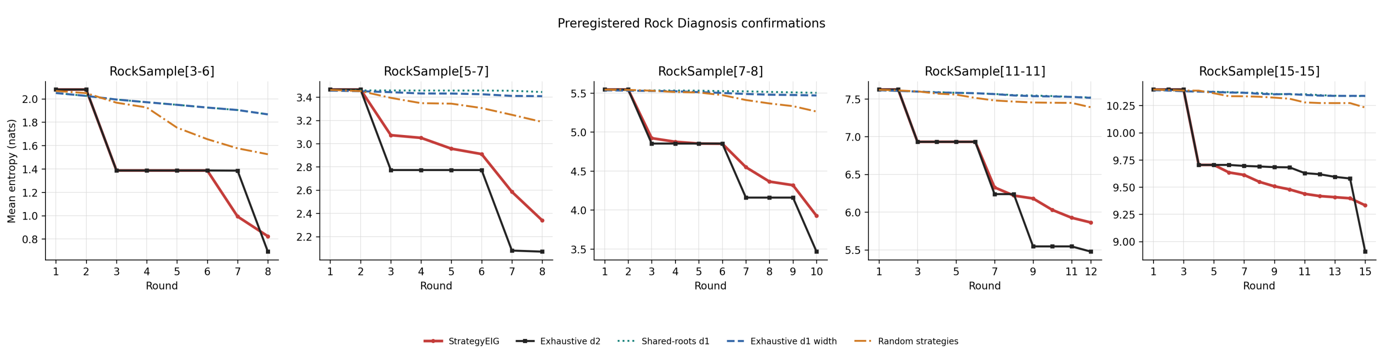

Figure 2: Mean posterior entropy over 30 paired trajectories per map. StrategyEIG and exhaustive d2 first spend zero-information actions on movement, then obtain large entropy drops from more accurate checks.

The mechanism is direct. StrategyEIG selects movement on 140/240 and 128/240 decisions on the smaller maps, 193/300 on RockSample[7,8], and 195/360 on RockSample[11,11], and 150/450 on RockSample[15,15], while shared d1 and exhaustive d1 width never move. Its proposals capture 90.6%, 86.1%, 85.8%, 73.9%, and 41.1% of exhaustive d2 value over depth-two rounds. Before the paid eight-rock scale extension, a 1,000-pair exact qualification found that exhaustive d2 reduced final entropy by

1.6775 nats relative to exhaustive d1, with 95% interval [1.6568, 1.6980]. Beating shared d1 shows that non-myopic scoring is load-bearing; beating matched random strategies shows that the LLM prior is load-bearing. The exhaustive d2 oracle remains better. A preregistered zero-call diagnostic closed each Gemma root over exact legal followups, raising mean exhaustive fraction from 85.8% to 99.6% and localizing most headroom to continuation choice. Optimal-root coverage was 242/270 (89.6%), one state below the frozen 90% paid-run gate, so we did not tune or run it.

Fresh machine-slot replications at eight rocks also passed all controls: GPT-5.4 Mini gained .5104, .4928, and .4130 entropy-AUC nats against shared d1, width, and random, while Gemma gained .7846, .7729, and .7334; every paired interval and truth-log corroboration excluded zero. Because seeds differ, cross-model effect sizes remain descriptive rather than causal model comparisons.

Finally, a 500-pair exact gate on RockSample[11,11] found that exhaustive d2 beat exhaustive d1 by 1.1042 entropy-AUC nats [1.1015, 1.1068]. Four independent Gemma seeds each passed all six entropy/truth gates. Across the three fresh robustness seeds, pooled entropy-AUC gains were .9470 [.9253, .9671], .9440 [.9228, .9637], and .8727 [.8359, .9073], with 90/0/0 each. GPT-5.4 Mini also passed all six, although its .5459 oracle gap shows model-dependent proposal quality. This extends the result to 2,048 latent states with zero rollout LLM calls. Exact closure raised LLM roots from 73.9% to 96.9% of oracle value, but random roots reached 96.7%, localizing the LLM advantage to continuation choice; the paid gate failed. Across the first four maps, K6 uses 13.7–14.2 scorer nodes/decision versus exhaustive d2's 64.3–354.2 (4.7×–24.9×). At 15 rocks, preregistered K4 uses 9.60 versus 616.94 nodes (64.3×). These are registered-budget tree-width, not wall-clock, ratios; K4 follows the 11-rock saturation result. A fresh 11-rock width sweep passed all 18 gates: K4 beat K2 by .5655 entropy-AUC nats [.4974, .6418], while K6 minus K4 was .0102 [-.0156, .0337]. K4 therefore matched K6 endpoint quality with 9.49 instead of 14.24 scorer nodes/decision.

A preregistered 100-pair exact qualification on the frozen RockSample[15,15] instance extends the structural mechanism to 32,768 latent vectors. Exhaustive d2 beat d1 by .5888 entropy-AUC nats [.5801, .5971] and .5930 truth-log-AUC nats [.5470, .6396], with 100/0/0 wins/ties/losses. A fresh direct-vLLM Gemma policy then passed all six registered gates: entropy-AUC gains were .6723 [.6328, .7124] against shared d1, .6700 [.6301, .7111] against exhaustive d1 width, and .6345 [.5722, .6877] against random strategies, with 30/0/0 each and positive truth-log intervals. Two fresh direct-vLLM seeds also passed all twelve gates; pooled 90-pair entropy-AUC gains were .6839 [.6626, .7051], .6815 [.6603, .7036], and .6760 [.6493, .7017] against shared d1, width, and random, with 90/0/0 each. A fresh GPT-5.4 Mini replication passed all six gates as well: .4716 [.3762, .5601], .4562 [.3653, .5482], and .4410 [.3245, .5483], respectively. It moved on 233/450 decisions; both myopic controls never moved, and rollout scoring made no LLM calls.

A preregistered smaller-proposer transfer exposed the corresponding capacity boundary. Non-thinking Gemma 4 E4B passed all legality and branch-mechanics checks, but its K4 proposals captured only 6.8% of exhaustive d2 value, versus 41.1–47.6% for the three 26B seeds. It beat shared d1 and random on entropy AUC, but failed the matched-width interval, .0408 [-.0035, .0884], and failed truth-log corroboration against shared d1 and width. Thus exact verification removes scoring noise but cannot recover a valuable branch absent from the bounded proposal set. A preregistered non-thinking 12B transfer passed all six gates, and two frozen fresh seeds independently replicated every entropy and truth-log interval. Across all 90 pairs, pooled entropy-AUC gains were .6417 [.6171, .6672], .6075 [.5813, .6340], and .6194 [.5930, .6451] against shared d1, width, and random, with 90/0/0 each. Its K4 support captured 49.5–53.3% of exhaustive d2 value across seeds. Different fresh seeds make the model-size contrast descriptive, but the replicated pass brackets a practical E4B-to-12B capacity boundary.

In a non-spatial Gated Sensor task, zero-EIG activation unlocks precise tests for a 32-state fault; exact d2 beat d1 by 1.2459 [1.2401,1.2516] entropy-AUC nats. The strongest GPT-5.4 Mini probe nevertheless tied random, −.0413 [−.1854, .0795], because its continuations redundantly activated at the terminal step. Verification cannot recover setup value absent from the proposal.

Three data-derived tasks reproduce the structural gap. On UCI Mushroom, exact d2 gained .1153 [.1057,.1254] over d1. Names-only 26B recovered .327 with 10/16 collection; grounded GPT collected 50/50 with zero projected branches, exactly matched d2, and gained .1023 [.0614,.1441] over d1 and .0685 [.0376,.1006] over random; truth-log intervals agreed. On UCI Cleveland, exact d2 gained .0687 [.0573,.0803] and ordered a workup 2.22 rounds earlier; 26B selected workup on 1/16 opportunities and lost to the strong d1 continuation. Both names-only gates failed. A fresh utility-grounded GPT-5.4 Mini gate instead selected workup 16/16 times and recovered all exact d2 opportunity. Over 50 paired eight-round trajectories it used zero projected branches, recovered 1.009 of exact d2, and gained .0631 [.0342,.0924] entropy AUC over d1 and .0289 [.0067,.0521] over matched random. Truth-log gain over d1 was positive; versus random, the registered interval was [.0001,.0629], but an independent bootstrap was [-.0007,.0623], so this is primary-endpoint transfer rather than a full-gate pass.

UCI Thyroid marks 16 variables immediate and TSH, T3, TT4, and T4U as one delayed blood-test group. On 1,000 fresh pairs, exact d2 collected then queried TSH, gaining .1659 [.1523,.1791] entropy AUC and .1669 [.1359,.1998] truth-log AUC over d1. A 26B proposal gate recovered .984 of d2, but trajectory serving failed. Names-only GPT-5.4 Mini gained .0641 [.0205,.1046] over d1, lost .0339 [-.0739,.0073] to matched-random continuations, and exposed ungrounded empirical assay utility; a utility-card repair fixed eight starts but failed before endpoints. A fresh projection-only ablation completed 50 pairs but gained only .0890 over d1, lost .0129 to random, recovered .432, and collected 0/50 with 2/3,765 branches projected. Fresh utility plus bounded legal projection changed only invalid branches after retry; entropy AUC gained .2026 [.1437,.2543] over d1 and .1185 [.0813,.1578] over random; truth-log intervals were positive, recovery was .994, and collection was 50/50. Only 17/4,338 branches were projected, supporting machine-grounded rather than unaided LLM planning.

More receding-horizon depth is not automatically better. In 500 exact pairs, terminal-EIG d3 lost to d2 by .2044 [.1998,.2090] entropy-AUC nats, 0/0/500: replanning repeated a noisy check and postponed its promised move. Endpoint weighting beat same-utility d2 by .2744 [.2689,.2799] but reproduced the prior d2 trajectory, creating no deeper oracle gain.

A range-gated variant made the extra step load-bearing: .55-accurate remote checks became .95 accurate on site, two moves away. Exact d3 beat exact d2 by .4630 [.4600,.4657] over 500 pairs. A successor-grounded 26B gate proposed the exact

South,South,check-5 optimum on 16/16 fresh cells, gaining .4809 [.4799,.4818] over identical-root random, .4859 [.4849,.4871] over shared d2, and .4796 over the strongest d2 root; exact-d3 recovery was 1.0. Replay matched all values, but two tied control identities differed across platforms. The frozen audit therefore failed and no trajectory ran: positive h3 proposal evidence only.

Paprika fails before a horizon claim.

Paprika troubleshooting mixes information-gathering questions with corrective actions that change and may terminate the world [Tajwar et al., 2025]. Pure EIG values discrimination among cause/remedy strings, not the probability of resolution minus action cost. A remedy that resolves under every hypothesis can have zero EIG, while a diagnostic that identifies the cause but cannot resolve the task can score highly.

The implementation also used fixed support and synthetic outcome labels in lookahead while deployment refreshed support, observed free text, and could terminate. A ten-task full-depth diagnostic used 70,004 requests versus 3,365 for one-step EIG and resolved 2 versus 3 tasks. Those counts are not policy evidence: later manual audit found a customer response claiming that the exact private remedy had failed. A separate 50-task arbitration evaluation was also quarantined by endpoint review. The defensible result is the failure mechanism, not which arm won an invalid simulator contest.

MediQ validates the interface but rejects the joint model.

MediQ's official FactSelect patient returns only atomic record facts that answer the question and otherwise abstains [Li et al., 2024]. After several failed closed repairs, the final five-case environment gate achieved clean canonical Yes/No/unavailable questions, verbatim grounding, relevance, and semantic deduplication on every selected interaction. This isolates the remaining issue to probability modeling.

The first scorer treated each A–D option as a complete patient world. Across ten selected turns, predicted EIG averaged 0.140 nats but realized true-label log-probability gain averaged -0.290 nats; the realized outcome was less likely under the true label than under the predictive mixture on 6/10 turns. A repair made record missingness label-independent and scored only Yes versus No under each option. Missingness became exactly posterior-neutral, yet available outcomes still had mean true-label log gain -0.118 nats.

A final data-estimation construction elicited the response marginal and hypothetical A–D posteriors, then projected their joint table to exact local marginals. Before scoring it, a frozen held-out bank required at least 15 available responses. All 30 questions were canonical, unique, grounded, and relevant, but only 8 were answerable; 22 facts were absent from the static records. The scorer was therefore not run. At observed answerability $a = 8/30$, an unavailable-neutral binary channel has the loose ceiling

$$I(\theta; Y \mid x) \leq a \log 2 = 0.1848 \text{ nats per query},$$

before accounting for diagnostic ambiguity. There is little signal on which a depth-two advantage could operate.

We then ran a fresh, preregistered iCRAFT-MD diagnosis-only profile gate. The official scorer uses a four-way diagnosis index and the records are denser, so we assigned three concrete counterfactual patient profiles to each diagnosis, held record availability label-independent, and froze support before any branch. The one-case micro-smoke passed: it produced all 12 profiles, two candidates, and canonical profile likelihood tables for \$0.0027. The original iCRAFT profile-support gate failed before likelihood when the 26B scaffold exhausted bounded profile construction on calibration. One separately authorized retry used a frontier model to author profiles only, with 26B retaining every other role. It reached calibration but only 14 of 48 grounded FactSelect outcomes were available, below the preregistered minimum of 24. This terminal channel-sparsity failure closes the path before structural, ranking, or policy evaluation; we made no prompt repair or rerun.

5 Why Horizon Magnifies Model Error

Let an estimated continuation score be $\widehat{V}_j = V_j + \epsilon_j$ for branch candidate j. Depth two uses $\max_j \widehat{V}_j$, so even zero-mean errors produce positive selection bias. Increasing branch width or horizon adds more opportunities for an optimistic likelihood, posterior, or proposal to win. Root likelihood error also changes branch mass and the posterior context passed to every downstream call. More Monte Carlo rollouts reduce variance conditional on this model; they do not reduce its bias. Bayesian design is particularly exposed to misspecification because the same model both predicts data and decides which data to collect [Catanach and Das, 2023].

BED-LLM reaches strong one-step results under cleaner conditions: a concrete entity or user profile, a small response space, a prior-likelihood construction, and diverse hypotheses retained with uniform mass after history filtering [Choudhury et al., 2026]. It explicitly reports raw in-context beliefs as overconfident and finds its data-estimation ablation weaker than prior-likelihood BED. Our MediQ belief similarly reached mean top probability 0.867 after round one at only 50% accuracy.

The Rock result instead follows the LLM-Modulo pattern: the model supplies a small, semantically informed policy set, while a sound external component checks grammar, legality, Bayesian updates, and total information gain [Kambhampati et al., 2024]. The comparison to random policies shows that the generator contributes useful search bias; the comparison to the same roots under d1 shows that external lookahead contributes the decision rule. Neither component alone explains the gain.

UoT reports depth gains on explicit possibility sets with only 5 or 15 diseases [Hu et al., 2024]. CA-BED uses finite exclusive hypotheses, soft LLM likelihoods, binary questions, and depth two [Arnould et al., 2026], but lacks a causal depth-one/depth-two ablation and its categorical extension hurts Detective. These results show structured planning can help, not that horizon survives an insufficient latent or incompatible answer categories.

Deep adaptive design and RL-BED optimize total sequential utility through amortized policies or reinforcement learning [Foster et al., 2021, Blau et al., 2022]. Both require trajectories from a meaningful simulator. Training on the current MediQ surrogate would amortize its annotation and calibration artifacts rather than recover a missing patient state.

6 A Validation-First Path Forward

Future depth studies should cross the following gates in order:

1. **Target and state:** the endpoint must decode θ, while a latent z must contain everything required to generate responses.
2. **Prior:** held-out log loss and Brier score must beat uniform, with no high-confidence wrong collapse.
3. **Likelihood:** available observations must improve mean true-target log probability; missingness must be neutral unless its semantics are genuinely target-dependent.
4. **Branch equivalence:** simulated and deployed histories must yield the same support, probabilities, and stopping behavior.
5. **Structural gap:** an exact or high-fidelity oracle must show depth two beating greedy under the benchmark's native action and budget rules.
6. **Ranking fidelity:** estimated incremental lookahead value must rank oracle incremental value under shared candidates and common random numbers.
7. **Policy test:** only then compare candidate horizons on paired cases, with identical root candidates, simulator seeds, and stopping rules.

7 Limitations and Conclusion

This is a validation study and positive structured-benchmark result, not yet a positive external-benchmark claim for non-myopic LLM BED. Rock Diagnosis uses an exact finite simulator; its diagnosis-only extensions omit reward, sampling, and exit. Rock Diagnosis does not test robustness to learned likelihoods or open responses. Its range-gated h3 result remains proposal-only: a tied-control identity mismatch failed the frozen cross-platform audit, so no trajectory ran. Mushroom and Cleveland use artificial unlocks and exact local summaries; Mushroom matched d2 without projection. Thyroid has native delayed/grouped metadata, but its 26B proposal pass failed trajectory serving and names-only GPT lost to random. Projected utility passed using exact local summaries and 17 projected branches, so it is not unaided LLM planning. Cleveland's grounded transfer likewise uses exact local summaries; it required no projection and passed the entropy endpoint, but its frozen independent truth-log interval against random crossed zero, so it is not an all-gates confirmation. Animals establishes one-step transfer, but its depth streams are only partially paired. Natural location uses one model family and environment. Paprika policy counts are endpoint-invalid, and MediQ stops before a calibrated policy comparison. We do not claim that non-myopic BED cannot work; the preregistered iCRAFT profile gate failed held-out calibration.

Capacity, one model family, OpenRouter provider nondeterminism, finite supports, unpaired model sizes, and post-hoc synthesis limit generality.

The evidence supports a constructive conclusion. With a coherent simulator and an enabling-information gap, LLM proposals plus exact rollout-EIG verification produce a large, paired, non-myopic gain; both generation and verification matter. But depth is not a substitute for a valid joint model, deployment-matched branches, or a task in which greedy information gain is genuinely suboptimal. Calibrated summaries plus bounded legal projection repaired Thyroid transfer, and the same summaries yielded a zero-projection full Mushroom transfer and primary-endpoint Cleveland transfer; projection-only failed random. These empirical gates should precede another non-monotonic depth sweep.

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